

Homework 3
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25.2:

1.

TimeId/Pid	11	12	13	Total
1	60	56	28	144
2	30	65	50	145
3	25	70	15	110
Total	115	191	93	399

2. SELECT S.pid, S.timeid, SUM(S.sales)
FROM Sales S
GROUP BY S.pid, S.timeid

This gets each of the sums of sales for each TimeId and Pid pair.

```
SELECT S.pid, SUM(S.sales)
FROM Sales S
GROUP BY S.pid
```

This gets the last row other than the bottom right element, the sums of sales for each Pid.

```
SELECT S.timeid, SUM(S.sales)
FROM Sales S
GROUP BY S.timeid
```

This gets the last column other than the bottom right element, the sums of sales for each TimeId.

```
SELECT SUM(S.sales)
FROM Sales S
```

This gets the bottom right element, the sum of sales across all Pids and TimeIds.

3.

LocId/Pid	11	12	13	Total
1	48	100	28	176

2	67	91	65	223
Total	115	191	93	399

25.10:

1. To decide whether to materialize a view, we need to consider:
 - The cost of refreshing the view
 - The cost of executing the view definition every time it is accessed (if it is very large, it's a good idea to materialize the view so this doesn't have to run)
 - The extra disk space used to store the materialized view
 - Whether it helps to build indexes on the materialized view
 - Whether refreshing the view will impact any other transactions
 - How often the view needs to be synchronized
2. a. An incremental view maintains the number of times a row is derived from the underlying relations. This allows for incremental adjustments to the view rather than having to recompute the entire view any time any of the underlying relations are updated.
 - b. Pros: You don't have to run the view definition query every time you want to access the view and you can index the materialized view making queries on it run faster
 - Cons: It's more difficult to update the materialized view, it takes more disk space, and the data may be outdated if the updates are not frequent enough
3. If you want a faster response time for the view you may want it at a different site than the underlying relations. The view could become inconsistent if the necessary updates do not occur before the view is queried.
4. a. It stores the information described in 2a which helps the algorithm which rows of the view will need to be updated when the underlying tables are updated.
 - b. Pros: In this case, it's helpful to not have to run the view definition query when you want to access the view because it contains a group by clause so it will be relatively time consuming
 - Cons: It's more difficult to update the materialized view, it takes more disk space, and the data may be outdated if the updates are not frequent enough

26.8:

- 1.

Age	Subscription = yes	Subscription = no
32	0	1
35	1	0
37	0	1
39	1	0
40	0	1
43	1	0
52	1	0
55	1	0
56	1	0

Salary	Subscription = yes	Subscription = no
43k	1	0
45k	0	1
50k	1	0
54k	0	1
58k	0	1
68k	1	0
70k	1	0
85k	1	0
90k	1	0

2. Child 1:

Age	Subscription = yes	Subscription = no
32	0	1
35	1	0

37	0	1
39	1	0
40	0	1
43	1	0

Salary	Subscription = yes	Subscription = no
45k	0	1
54k	0	1
58k	0	1
68k	1	0
70k	1	0
90k	1	0

Child 2:

Age	Subscription = yes	Subscription = no
52	1	0
55	1	0
56	1	0

Salary	Subscription = yes	Subscription = no
43k	1	0
50k	1	0
85k	1	0

26.10:

- $\text{norm}((1, 3, 4) - (2, 3, 2)) = \sqrt{(1-2)^2 + (3-3)^2 + (4-2)^2} = \sqrt{5} = 2.236$
- $\text{norm}((1, 3, 4) - (3, 3, 7)) = \sqrt{(1-3)^2 + (3-3)^2 + (4-7)^2} = \sqrt{13} = 3.605$

c. $\text{norm}((3, 3, 7) - (2, 3, 2)) = \sqrt{(3-2)^2 + (3-3)^2 + (7-2)^2} = \sqrt{26} = 5.099$

Question 21.2

1. Show the view definition statements for Employee Names and DeptInfo.

```
CREATE VIEW EmployeeNames (ename) AS
    SELECT E.ename
    FROM Employees E
```

```
CREATE VIEW DeptInfo (dept, avgsalary) AS
    SELECT E.dept, AVG(E.salary) AS avgsalary
    FROM Employees E
    GROUP BY E.dept
```

2. A user who needs to know only the department salaries for the Toy and CS departments should be granted SELECT privilege on the view DeptInfo because it will be difficult to view the average for only these two departments.
3. The privileges that should be granted to the secretary are:
 - DELETE on Employess (to fire people)
 - SELECT on EmployeeNames view (to check who is an employee)
 - SELECT on DeptInfo view (to check on average department salaries)
4. Based on the privileges given above, it is not possible for the secretary to find out the salary of an employee using the relations alone but if there is just one employee in a department and the secretary knows about this along with the name of the employee who works there, then he can possibly find out the salary.
5. The command to give the secretary command to allow other people to read the EmployeeNames is:

```
GRANT SELECT ON EmployeeNames TO Secretary WITH GRANT OPTION
```

6. The command to give the secretary right to insert tuples into EmployeeNames view is:

```
GRANT INSERT ON Employees TO Secretary
```

With this command, the secretary has right to insert tuples into the AtoR view since its updatable but not the HowManyNames view as it cannot be updated. The secretary will still not have the SELECT privilege.

7. Todd privileges will also be revoked considering it was given by the secretary. If other users had given Todd privileges, then he will retain those ones.

8. The average salary for a department cannot be changed as for that to be implemented, a person will need to know which salaries to change which is impossible.

9. The commands to give Joe authorization to read and modify the Employees and EmployeesNames relations are:

```
GRANT SELECT, UPDATE, INSERT ON Employees TO Joe WITH GRANT  
OPTION
```

```
GRANT SELECT, UPDATE, INSERT ON EmployeeNames TO Joe WITH  
GRANT OPTION
```

And this command doesn't automatically give Joe privilege to read DeptInfo.

10. You cannot revoke Mike rights without revoking Joe rights in SQL even temporarily.

11. To revoke Joe's right to prevent Mike and Susan from accessing the view, the revoke statement will be:

```
REVOKE SELECT ON EmployeeNames FROM Joe
```

This statement will revoke Joe's right to read EmployeeNames, and drop the AllNames view. Joe will still be able to modify EmployeeNames but cannot read records.

Question 25.4

1. The difference between WINDOW clause and the GROUP BY clause is that WINDOW clause is faster as it utilizes the WINDOW SORT operation on data that has

already been fetched by a query while the GROUP BY clause utilizes the HASH GROUP BY operation on data taken directly from the table.

2. Rank all universities in the state based on their budget from the previous year and for each university, print the difference in budgets between them and the university ranked behind it.

3. In SQL:1999, the frame of a window is the boundaries of the window associated with each row in terms of the ordering of rows of partitions.

4. The query in SQL:1999 will be:

```
SELECT T.year, SUM(S.sales) OVER W AS movsum
FROM Sales S, Times T
WHERE S.timeid = T.timeid
WINDOW W AS( PARTITION BY T.year ORDER BY T.year)
```

Question 26.2

1. The support of item sets of size 1 are:
 - {pen} = 100%

- {ink} = 75%
- {milk} = 75%
- {juice} = 50%
- {water} = 25%

The only itemset that meets the minimum support threshold of 90% is the {pen} itemset and no other one element itemset. Because there is no more one element itemset that meets the minimum support threshold of 90% we can conclude that there does not exist any item sets of size 2 that do so.

There are no association rules that meet the 90% minimum confidence threshold because there only exists one one element itemset that meets the minimum support threshold

2. One example possibility is shown below where support for the items {ink} and {milk} is reduced to below 75%.

Transid	CustID	Date	Item	Qty
111	201	5/1/99	Pen	2
111	201	5/1/99	Milk	3
111	201	5/1/99	Juice	6
112	105	6/3/99	Pen	1
112	105	6/3/99	Ink	1
113	106	5/10/99	Pen	1
113	106	5/10/99	Milk	1
114	201	6/1/99	Pen	2
114	201	6/1/99	Ink	2
114	201	6/1/99	Juice	4
114	201	6/1/99	Water	1

3. All the item sets of size 1 meet the minimum support threshold of 10%

- {pen} = 100%
- {ink} = 75%
- {milk} = 75%
- {juice} = 50%
- {water} = 25%

All the item sets of size 2 meet the minimum support threshold of 10% other than {milk, water}

- {pen, ink} = 75%
- {pen, milk} = 75%

$\{\text{pen, juice}\} = 50\%$
 $\{\text{pen, water}\} = 25\%$
 $\{\text{ink, milk}\} = 50\%$
 $\{\text{ink, juice}\} = 50\%$
 $\{\text{ink, water}\} = 25\%$
 $\{\text{milk, juice}\} = 25\%$
 $\{\text{milk, water}\} = 0\%$
 $\{\text{juice, water}\} = 25\%$

Using the ones with size 2 that meet the support threshold, the item sets of size 3 to consider are:

$\{\text{pen, ink, milk}\} = 50\%$
 $\{\text{pen, ink, juice}\} = 50\%$
 $\{\text{pen, ink, water}\} = 25\%$
 $\{\text{pen, milk, juice}\} = 25\%$
 $\{\text{pen, juice, water}\} = 25\%$
 $\{\text{ink, milk, juice}\} = 25\%$
 $\{\text{ink, juice, water}\} = 25\%$

Using the ones with size 3 that meet the support threshold, the item sets of size 4 to consider are:

- $\{\text{pen, ink, milk, juice}\} = 25\%$
- $\{\text{pen, ink, juice, water}\} = 25\%$

There can't be any item sets of size 5 meeting the threshold because $\{\text{milk, water}\}$ has 0 support.

So, these are all the itemsets that meet the threshold:

- $\{\text{pen}\} = 100\%$
- $\{\text{ink}\} = 75\%$
- $\{\text{milk}\} = 75\%$
- $\{\text{juice}\} = 50\%$
- $\{\text{water}\} = 25\%$
- $\{\text{pen, ink}\} = 75\%$
- $\{\text{pen, milk}\} = 75\%$
- $\{\text{pen, juice}\} = 50\%$
- $\{\text{pen, water}\} = 25\%$
- $\{\text{ink, milk}\} = 50\%$
- $\{\text{ink, juice}\} = 50\%$

- {ink, water} = 25%
- {milk, juice} = 25%
- {juice, water} = 25%
- {pen, ink, milk} = 50%
- {pen, ink, juice} = 50%
- {pen, ink, water} = 25%
- {pen, milk, juice} = 25%
- {pen, juice, water} = 25%
- {ink, milk, juice} = 25%
- {ink, juice, water} = 25%
- {pen, ink, milk, juice} = 25%
- {pen, ink, juice, water} = 25%

These association rules have confidence $\geq 90\%$

- {ink} $\rightarrow$ {pen}
- {milk} $\rightarrow$ {pen}
- {juice} $\rightarrow$ {pen}
- {juice} $\rightarrow$ {ink}
- {juice} $\rightarrow$ {pen, ink}
- {water} $\rightarrow$ {pen}
- {water} $\rightarrow$ {ink}
- {water} $\rightarrow$ {juice}
- {water} $\rightarrow$ {pen, ink}
- {water} $\rightarrow$ {pen, juice}
- {water} $\rightarrow$ {ink, juice}
- {water} $\rightarrow$ {pen, ink, juice}
- {ink, milk} $\rightarrow$ {pen}
- {ink, juice} $\rightarrow$ {pen}
- {ink, water} $\rightarrow$ {pen}
- {ink, water} $\rightarrow$ {juice}
- {juice, water} $\rightarrow$ {ink}
- {juice, water} $\rightarrow$ {pen}
- {milk, juice} $\rightarrow$ {pen}
- {milk, juice} $\rightarrow$ {ink}
- {pen, juice} $\rightarrow$ {ink}
- {pen, water} $\rightarrow$ {ink}
- {pen, water} $\rightarrow$ {juice}
- {milk, juice} $\rightarrow$ {pen, ink}
- {pen, water} $\rightarrow$ {ink, juice}
- {ink, water} $\rightarrow$ {pen, juice}
- {juice, water} $\rightarrow$ {pen, ink}

4. Currently the support is defined as follows:

{pen} - 4/4

- {ink} - 3/4
- {milk} - 3/4
- {pen, ink} - 3/4
- {pen, milk} - 3/4
- {ink, milk} - 2/4

Any of these will work because they don't decrease the support of {pen,milk} or {pen,ink}:

- {111, 201, 5/1/99, pen, 2}
- {112, 105, 5/1/99, pen, 2}
- {113, 106,5/10/99, pen, 1}
- {113, 106,5/10/99, ink, 1}
- {114, 201, 6/1/99, milk, 2}
- {114, 201, 6/1/99, pen, 2}

Question 26.9

1. Assuming all six records belong to a single cluster:

Given records: (7,55), (21, 202), (25,220), (12,73), (8,61), and (22,249).

Center (C1):

$$C1 = \{7+21+25+12+8+22\}/\{6\} = 15.833, \{55+202+220+73+61+249\}/\{6\} = 143.33$$

Therefore the Center is (15.833, 143.33),

Radius (R1):

$$R1 = \max(\sqrt{(7-C1_x)^2 + (55-C1_y)^2}, \sqrt{(21-C1_x)^2 + (202-C1_y)^2}, \dots) = \sqrt{38.03 + 11166.15} = 105.85$$

The Radius is ≈ 105.8465

2. Assuming the first three records belong to one cluster and the second three records belong to a different cluster:

Given records: (7,55), (21,202), (25,220), (12,73), (8,61), and (22,249).

Center (C2) for the first cluster:

$$C2_1 = (\{7+21+25\}/\{3\}, \{55+202+220\}/\{3\}) = 17.67, 159$$

Radius (R2_1):

$$R2_1 = \max(\sqrt{(7-C2_1_x)^2 + (55-C2_1_y)^2}, \sqrt{(21-C2_1_x)^2 + (202-C2_1_y)^2}, \dots) = \sqrt{113.85+10816} = 104.55$$

Center (C2) for the second cluster:

$$C2_2 = (\{12+8+22\}/\{3\}, \{73+61+249\}/\{3\}) = 14, 127.67$$

Radius (R2_2):

$$R2_2 = \max(\sqrt{(12-C2_{2_x})^2 + (73-C2_{2_y})^2}, \sqrt{(8-C2_{2_x})^2 + (61-C2_{2_y})^2}, \dots) = \sqrt{64+14720.97} = 121.6$$

Therefore:

The center for the first cluster is (17.67, 159) and the Radius≈104.5456

The last Center for the second cluster is (14,127.67) and the Radius≈121.5968

3. The first cluster is better than the second cluster based on the maximum radius